One-dimensional Kalman filters: review

Labbe, Kalman and Bayesian Filters in Python, ch. 4, pp. 131–170

Lesson t2.2 · R_proc is Labbe's Q, Q_meas is Labbe's R

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Review order

1. Predict (C1)
2. Update, Bayes form (C2)
3. Update, gain form and loop (C3)
4. Steady state and fixed gain (C4)
5. Design traps
6. Mistakes already made

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Names: Labbe versus this package

Labbe	Package	Meaning
Q, process_var	R_proc	variance added per predict
R, sensor_var	Q_meas	measurement variance
P, x.var	P ($\bar{P}$ prior)	belief variance
dx, u, μ_fx	u	known displacement per step

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Labbe's Q and R are the opposite of Thrun's; the package never writes a bare R or Q

Predict, pp. 135–136 and 153

$$\bar{x} = x + u, \qquad \bar{P} = P + R_{proc}$$

p. 153

x, x̄ mean, m

P, P̄ variance, m²

u m/step

R_proc m²/step

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Trap: R_proc = 0 because u is exact; the filter stops listening

C1 robust predict

```
def predict(x, P, u, R_proc, dt=1.0):
    assert P >= 0.0 and R_proc >= 0.0
    return x + u * dt, P + R_proc * dt
```

- Separate floats, named for what they are: FilterPy's signature with Q renamed
- Scale R_proc with Δt; a dropped measurement is just another predict

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Update, Bayes form, p. 138

$$\mu = \frac{\bar{\sigma}^2 \mu_z + \sigma_z^2 \bar{\mu}}{\bar{\sigma}^2 + \sigma_z^2}, \qquad \sigma^2 = \frac{\bar{\sigma}^2 \sigma_z^2}{\bar{\sigma}^2 + \sigma_z^2}$$

p. 138

$\bar{\mu}$, $\bar{\sigma}^2$ prior

μ_z, σ_z^2 z and Q_meas

μ, σ^2 posterior

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Trap: σ_z from a datasheet where σ_z² is wanted

C2 naive update, p. 138

```
def gaussian_multiply(g1, g2):
    mean = (g1.var*g2.mean + g2.var*g1.mean) / (g1.var + g2.var)
    variance = (g1.var * g2.var) / (g1.var + g2.var)
    return gaussian(mean, variance)
```

- Robust form: precisions add, and total ignorance is $\bar{P} = \infty$, not 10^6
- The product's constant $N(z; \bar{x}, \bar{P} + Q_{meas})$ is the measurement likelihood

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Update, gain form, p. 153

$$y = z - \bar{x}, \quad K = \frac{\bar{P}}{\bar{P} + Q_{meas}}, \quad x = \bar{x} + Ky, \quad P = (1 - K)\bar{P}$$ p. 153

- **y** residual, m
- **K** gain
- **S** P̄ + Q_meas

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Trap: the book's chart label reads y = z − −x̄; it means z − x̄

C3 gain-form update, p. 151

```
def update(prior, measurement):
    x, P = prior
    z, R = measurement      # Labbe's R is Q_meas
    y = z - x
    K = P / (P + R)
    x = x + K*y
    P = (1 - K) * P
    return gaussian(x, P)
```

- Robust form: $S = \bar{P} + Q_{meas}$ must be positive; $P = \bar{P}\, Q_{meas} / S$ avoids $1 - K$
- Gate on y^2/S against the chi-square bound 6.635

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C3 the loop, p. 147

```
for z in zs:
    dx = velocity*dt
    pos = pos + dx
    var = var + process_var
    pos = (var*z + sensor_var*pos) / (var + sensor_var)
    var = (var * sensor_var) / (var + sensor_var)
```

- process_var is R_proc; sensor_var is Q_meas
- The two update lines are the Bayes form; the gain form is the same filter

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Bayes form or gain form

Bayes form (loop, p. 147)

- Weights prior and z by variances
- Shows the product directly

Gain form (p. 151)

- Prior plus K times residual
- Generalizes to Thrun's Table 3.1

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Same filter; the gap between them is float cancellation in the variance

What breaks first, card by card

Card	Breaks first	Then
C1 predict	R_proc too small (p. 160)	Biased u, irregular Δt
C2 Bayes update	$\bar{P} = 0$ and Q_meas = 0: 0/0	float32 with huge $\bar{P}$; σ for σ^2
C3 gain form	R_proc too small; NIS ⩾ 1	Datasheet Q_meas; no gate
C4 fixed gain	Noise changes, K does not	Transient; biased u

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First step and steady state, measured

401

first prior variance

1.990

1

first posterior variance

2.162

3

P∞ for (2, 4.5)

0.480

5

K∞ for (2, 4.5)

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Steady state

$$P_\infty^2 + R_{proc}P_\infty - R_{proc}Q_{meas} = 0, \qquad K_\infty = \frac{P_\infty}{Q_{meas}}$$

not in the chapter

P∞ fixed point of the posterior variance

K∞ steady-state gain

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Trap: the book's 0.13232 matches none of the chapter's filters

Steady state of each filter in the chapter

Filter	Page	P∞	K∞
Dog, 10 steps	143	1.00000	0.50000
Dog, 25 steps	147	2.16228	0.48051
Thermometer	156	0.00537	0.31770
Extreme noise	159	423.26525	0.00470
Bad process variance	160	0.14092	0.00705
Bad initial estimate	161	6.14143	0.24566
Large noise, bad init	162	243.95102	0.00813

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The book's 1.0, 2.1623 and 0.005 agree; 0.13232 on p. 168 does not

C4 fixed-gain update, p. 168

```
K = .13232          # the book's experimentally derived gain
y = z - x
x = x + K * y
```

- The book's K is not any chapter filter's K∞; recompute K∞ from R_proc and Q_meas
- Recompute when the loop rate changes: R_proc is per step

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Process variance: right model, wrong model

Right model (p. 159)

- $\sigma_z = 300$ m, $R_{proc} = 2$ m²
- RMS 15.06 m against 292.8 m raw

Wrong model (p. 160)

- $R_{proc} = 0.001$ on an accelerating dog
- Mean NIS 342.1 against 1.82

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P and K never see the data; NIS does

Symptom to fix

Symptom	Cause	Try
Drifts off data; NIS $\gg 1$	R_proc too small	Raise R_proc until NIS ≈ 1
K decays toward 0	R_proc = 0	Small nonzero R_proc
As noisy as z; NIS $\ll 1$	R_proc large or Q_meas small	Measure Q_meas at rest
Constant offset	Biased u	Fix u or add a bias state
Lags, flattens peaks	No dynamics	Larger R_proc or velocity
Long transient	Bad x_0	$x_0 = z_0$, P_0 = Q_meas
Off by a square	σ for σ^2	Square σ; name vars *_var

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Mistakes already made (REVIEW.md)

- The book does name the source of K = 0.13232: the dog tracking filter, p. 167
- The book uses `sensor_noise` as a variance on p. 165; only the name misleads
- $\bar{P}$ = 401 is on pp. 144 and 146, not p. 143, which holds only code
- The exact quote is the filter does not 'notice' that it is diverging, p. 161
- The nonlinear exercise has three effects: wrong u, lag, sensor variance 30 against 1.44

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The one thing to keep

R_proc is trust in the model, Q_meas is trust in the sensor; only their ratio sets K∞

Log K and NIS on every scalar filter

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